

IS 302-2-2 (1997)

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भारतीय मानक

घरेलू और समान विद्युत साधित्रों की सुरक्षा

भाग 2 प्रमुख अपेक्षाएँ

अनुभाग 2 निर्वात मार्जक और जल खूषण मार्जन उपकरण

Indian Standard

**SAFETY OF HOUSEHOLD AND
SIMILAR ELECTRICAL APPLIANCES**

PART 2 PARTICULAR REQUIREMENTS

Section 2 Vacuum Cleaners and Water Suction Cleaning Appliances

ICS 97.080

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**BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG**

NEW DELHI 110002

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Electrical Appliances Sectional Committee, ETD 32

FOREWORD

This Indian Standard (Part 2/Section 2) was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council

This Indian Standard covers the general and safety requirements of vacuum cleaners and water suction cleaning appliances. This standard does not cover the performance requirements, which will be covered under a separate standard at a later date.

This standard is to be read in conjunction with IS 302-1(1979) 'Safety of household and similar electrical appliances. Part 1 General requirements'. For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979), instead of reproducing full text of each clause, clauses of IS 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable or not applicable, the wording used is This clause of IS 302-1 (1979) is applicable/not applicable.
- b) In case of sub-clause or part thereof 'Not applicable'.

Wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

'Replacement — followed by the new text'.

Any addition to the existing provisions of a sub-clause of IS 302-1(1979) has been indicated as under:

'Addition — followed by the text of the additional matter'.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101 and additional sub-clauses are numbered with the main clauses number followed by 101, 102 etc. for example, 7.101.

Should however, any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance has been derived from the following publications issued by the International Electrotechnical Commission:

- IEC Pub 335-1 (1976) 'Safety of household and similar electrical appliances: Part 1 General requirements' and
- IEC Pub 335-2-2 (1993) 'Safety of household and similar electrical appliances: Part 2 Particular requirements for vacuum cleaners and water suction cleaning appliances'.

The Indian Standard which are necessary adjuncts to this standard are given in Annex AA.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2:1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 2 Vacuum Cleaners and Water Suction Cleaning Appliances

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows:

1.1 Replacement

This standard deals with the safety of electric vacuum cleaners and water suction cleaning appliances for household and similar purposes, including vacuum cleaners for animal grooming, their rated voltage being not more than 250 V.

This standard also applies to motorized cleaning heads and current-carrying hoses associated with a particular vacuum cleaner.

Appliances not intended for normal household use but which nevertheless may be a source of danger to the public such as appliances intended to be used by laymen in shops and other premises for normal housekeeping purposes are within the scope of this standard.

NOTES

1 Examples are appliances intended to be used for normal housekeeping purposes in hotels, offices, schools, hospitals and similar premises.

So far as is practicable, this standard deals with the common hazards presented by appliances which are encountered by all persons in and around the home.

This standard does not in general take into account:

The use of appliances by young children or infirm persons without supervision;

Playing with the appliance by young children.

2 Attention is drawn to the fact that:

For appliances intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary.

3 This standard does not apply to:

Appliances incorporating heating elements.

Centrally-sited vacuum cleaners;

Appliances designed exclusively for industrial purposes;

Appliances intended to be used in locations where special conditions prevail such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas).

2 TERMINOLOGY

2.1 This clause of IS 302-1 (1979) is applicable except as follows:

2.4 Addition

NOTE For appliances incorporating a booster setting the rated power input corresponds to the operation of the appliance when the booster setting is not activated

2.29 Replacement

Normal operation — The appliance is supplied at rated voltage and operated continuously with the air inlet adjusted to give a power input P_m after 20 s. Three min later a final adjustment of the air inlet is made, if necessary.

P_m is calculated from the following formula

$$P_m = 0.5 (P_f + P_i)$$

where

P_f is the power input in watts, after a min of operation with the air inlet unobstructed. Any device which ensures a flow of air to cool the motor in the event of a blockage of the main air inlet is allowed to operate:

P_i is the power input in watts, after a further 20 s of operation with the air inlet blocked. Any device which ensures a flow of air to cool the motor in the event of a blockage of the main air inlet is rendered inoperative

If the appliance is marked with a rated voltage range, it is supplied at the mean value of the range if the difference between the limits of the range does not exceed 10 percent of the mean value. If the difference exceeds 10 percent the supply voltage is the upper value of the range.

The measurements are made with the appliance fitted with a clean dust bag and filter, any water collection container being empty. If the appliance is intended for use only with a hose, detachable nozzles and tubes are removed and the hose is laid out straight. If the appliance is provided with a hose as an accessory, it is operated without the hose.

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Rotating brushes and similar devices are in operation but not in contact with any surface. Motorized cleaning heads are connected by means of the hose and are in operation but not in contact with any surface.

Appliance outlets for other accessories are loaded with a resistive load in accordance with the marking.

2.101 Water Suction Cleaning Appliances

Appliance for aspirating an aqueous solution which may contain a foaming detergent.

2.102 Booster Setting

Position of a control resulting in a temporary higher power input which is automatically reduced to that obtained when the setting is not used.

3 GENERAL REQUIREMENT

This clause of IS 302-1 (1979) is applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable except as follows:

4.2 Additional

A new hose is used for each of the tests of **21.101** to **21.104**.

4.101 Current-carrying hoses operating at safety extra-low voltage are not subjected to the tests.

5 RATING

This clause of IS 302-1 (1979) is applicable.

6 CLASSIFICATION

This clause of IS 302-1 (1979) is applicable except as follows:

6.1 Replacement

Vacuum cleaners and water suction cleaning appliances shall be of Class I, Class II or Class III.

Vacuum cleaners for animal grooming shall be of Class II or Class III.

Compliance is checked by inspection and by the relevant tests.

6.2 Addition

Vacuum cleaners for animal grooming and water suction cleaning appliances shall be at least splash proof.

7 MARKING

This clause of IS 302-1 (1979) is applicable except as follows:

7.1 Addition

Appliance outlets for accessories shall be marked with the maximum load in watts

NOTE This marking may be on the appliance close to the appliance outlet.

The sum of the rated power input of the vacuum cleaner and the maximum load of the appliance outlet shall also be marked on the appliance.

Motorized cleaning heads shall be marked with

- rated voltage or rated voltage range in volts.
- rated power input in watts,
- name, trade-mark or identification mark of the manufacturer or responsible vendor.
- model or type reference.

7.12 Addition

The instructions for use for appliances provided with a current-carrying hose operating at other than safety extra-low voltage shall include the substance of the following:

CAUTION This hose contains electrical connections.

- Do not use to suck up water,
- Do not immerse in water for cleaning.
- The hose should be checked regularly and must not be used if damaged.

7.12 Addition

The instructions for vacuum cleaners incorporating rotating brushes or similar devices and water suction cleaning appliances shall state that the plug must be removed from the socket-outlet before cleaning or maintaining the appliance.

7.101 The vacuum cleaner and water suction cleaning appliance may also be marked with Standard Mark

7.101.1 The use of Standard Mark is governed by the provisions of *Bureau of Indian Standards Act, 1986* and the rules and regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

8 PROTECTION AGAINST ACCESS TO LIVE PARTS

This clause of IS 302-1 (1979) is applicable.

8.1 Addition

If the instructions state that a part is to be removed

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when replacing a lamp or a drive belt and a tool is required for its removal the part is not considered to be a detachable part provided that:

- An instruction to disconnect the appliance from the supply before the opening is marked on the cover or is visible during its removal, and
- After removal of the cover, access to live parts is prevented by at least basic insulation.

9 STARTING OF MOTOR-OPERATING APPLIANCES

This clause of IS 302-1 (1979) is applicable.

10 INPUT

This clause of IS 302-1 (1979) is applicable except as follows:

10.1 Addition

The power input of motorized cleaning heads is checked separately.

NOTE — Appliance outlets are not loaded when measuring the rated power input.

Booster Setting are not used during these measurements.

11 TEMPERATURE RISE

This clause of IS 302-1 (1979) is applicable except as follows:

11.3 Addition

NOTE — When measuring the power input to ensure that the appliance has been correctly reassembled, the power input P_1 with the air-inlet blocked, is measured.

11.5 Addition

Booster Setting are activated as often as allowed by the construction.

11.7 Addition

Appliances are operated until steady conditions are established.

Appliances incorporating an automatic cord reel are operated with one-third of the total length of the cord unreeled for 30 min after which the cord is completely unreeled.

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1 (1979) is not applicable.

13 LEAKAGE CURRENT AND ELECTRIC STRENGTH AT OPERATING TEMPERATURE

This clause of IS 302-1 (1979) is applicable except

as follows:

13.1 Addition

NOTE The booster setting is not activated

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is not applicable

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows:

15.3 Replacement

Appliances having a liquid container shall be constructed so that spillage of liquid due to overfilling and due to overturning of unstable appliances and hand-held appliances does not affect their insulation

NOTE An appliance is considered to be unstable ??? overturns when a force of 180 N is applied to the top of the appliance in the most unfavourable horizontal direction The appliance is placed on a support inclined at an angle of 10° to the horizontal the liquid container being filled to half the level indicated in the instructions for use

Compliance is checked by the following test

Appliances with type X attachment except those having a specially prepared cord are fitted with the lightest permissible type of flexible cord of the smallest cross-sectional area specified in 26.2 of IS 302-1 (1979)

Appliances incorporating an appliance inlet are tested with or without an appropriate connector in position, whichever is most unfavourable.

Liquid containers which are filled by hand are completely filled with water containing approximately 1 percent NaCl and a further quantity, equal to 15 percent of the capacity of the container or () 25 l whichever is the greater, is poured in steadily over a period of 1 min

Containers of hand-held appliances and unstable appliances are completely filled the cover being closed The appliance is then overturned and left in that position for 5 min, unless it returns automatically to its normal position of use

The nozzle of water suction cleaning appliances is placed in a container, the base of which is level with the surface supporting the appliance The container is filled with a detergent solution to a level of 5 mm above its base, this level being maintained throughout the test. The solution consists of 20 g of NaCl and 1 ml of a solution of 28 percent by mass of dodecyl sodium sulphate in each 8 litre of water.

The appliance is operated until its liquid container is completely full and for a further 5 min.

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NOTE — The solution is to be stored in a cool atmosphere and used within seven days of its preparation. The chemical designation of dodecyl sodium sulphate is $C_{12}H_{25}NaSO_4$.

After each of these tests, the appliance shall withstand the electric strength test of **16.4**. However, for water suction cleaning appliances, the test voltage is:

- 1 000 V for basic insulation.
- 2 750 V for supplementary insulation.
- 3 750 V for reinforced insulation.

Inspection shall show that there is no trace of liquid on insulation which could result in a reduction of creepage distances and clearances below the values specified in **29.1**.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (AFTER HUMIDITY TREATMENT)

This clause of IS 302-1 (1979) is applicable except as follows:

16.4 Addition

Current-carrying hoses, except for their electrical connections, are immersed for 1 h in water containing approximately 1 percent NaCl, at a temperature of $27^{\circ}\text{C} \pm 5^{\circ}\text{C}$. While the hose is still immersed, a voltage of 2 000 V is applied for 5 min between each conductor and all the others connected together. A voltage of 3 750 V is then applied for a 1 min between all the conductors and the water

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable.

18 ENDURANCE

This clause of IS 302-1 (1979) is not applicable.

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows:

19.1 Addition

The test of **19.6** is only made on motorized cleaning heads.

Water suction cleaning appliances provided with a valve are additionally subjected to the test of **19.101**.

Appliances incorporating a booster setting which is not deactivated electronically, are additionally subjected to the test of **19.102**.

19.6 Addition

Motorized cleaning heads are tested with the rotating brush or similar device locked for 30 s.

19.8 Not Applicable

19.10 Replacement

Appliances incorporating series motors are supplied at 1.3 times rated voltage and operated for 30 s with the air inlet blocked, brushes and similar devices being removed.

After this test, the safety of the appliance shall not have been impaired, in particular windings and connections shall not have worked loose.

19.101 Water suction cleaning appliances, the container of which is provided with a valve or other protective device, are subjected to the test with the nozzle placed in a container as specified in **15.3** but with the valve or protective device held open or otherwise rendered inoperative.

NOTE If the appliance incorporates more than one protective device, these are rendered inoperative in turn

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows:

20.1 Addition

Covers, lids and accessories are placed in the most unfavourable positions.

For the purpose of this test any spillage of liquid is ignored.

20.2 Not Applicable

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable except as follows:

21.101 Current-carrying hoses shall be resistant to crushing.

Compliance is checked by the following test.

The hose is placed between two parallel steel plates each having a length of 100 mm, a width of 50 mm and the edges of the longer sides rounded with a radius of 1 mm.

The axis of the hose is positioned at right angles to the longer sides of the plates. The plates are placed at a distance of approximately 350 mm from one end of the hose.

The steel plates are pressed together at a rate of 50 ± 5 mm/min until the applied force is 1.5 kN.

The force is then released and the electric strength test of **16.4** is carried out between the conductors

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and metal foil applied to the outside of the hose, the test voltage however being 2 500 V.

21.102 Current-carrying hoses shall be resistant to abrasion.

Compliance is checked by the following test.

One end of the hose is attached to the connecting rod of the crank mechanism shown in Fig. 101. The crank rotates at 30 rev/min which results in the end of the hose moving horizontally backwards and forwards over a distance of 300 mm.

The hose is supported by a rotating smooth roller over which a belt of abrasive cloth moves at a speed of 0.1 m/min. The abrasive is corundum grit size P 100 as specified in IS 11986 1987.

A mass of 1 kg is suspended from the other end of the hose which is guided to avoid rotation. In the lowest position, the mass has a maximum distance of 600 mm from the centre of the roller.

The test is carried out for 100 revolutions of the crank.

After the test, basic insulation shall not be exposed and the electric strength test of **16.4** is carried out between the conductor and metal foil applied to the outside of the hose, the voltage however being 1 250 V.

21.103 Current-carrying hoses shall be resistance to flexing.

Compliance is checked by the following test.

The end of the hose intended to be connected to the motorized cleaning head is attached to the pivoting arm of the test equipment shown in Fig. 102. The distance between the pivot axis of the arm and the point where the hose enters the rigid part is 300 mm $\pm$ 5 mm. The arm can be raised from the horizontal position by an angle of $40^\circ \pm 1^\circ$. A mass of 5 kg is suspended from the other end of the hose or from a convenient point along the hose so that when the arm

is in the horizontal position the mass is supported and there is no tension on the hose

NOTES

1 It may be necessary to reposition the mass during the test.

The mass slides against an inclined plate so that the maximum deflection of the hose is 3° .

The arm is raised and lowered by means of a crank which rotates at a speed of 10 rev/min $\pm$ 1 rev/min.

The test is carried out for 2 500 revolutions of the crank after which the fixed end of the hose is turned through 90° and the test continued for a further 2 500 revolutions. The test is repeated in each of the other two 90° positions.

2 If the hose ruptures before 10 000 revolutions of the crank, the test is terminated.

After the test, the hose shall withstand the electric strength test of **16.4**.

21.104 Current-carrying hoses shall be resistance to torsion.

Compliance is checked by the following test.

One end of the hose is held in a horizontal position with the remainder of the hose freely suspended. The end is rotated in cycles, each cycle consisting of five turns in one direction and five turns in the opposite direction, at a rate of 10 turns per min.

The test is carried out for 2 000 cycles.

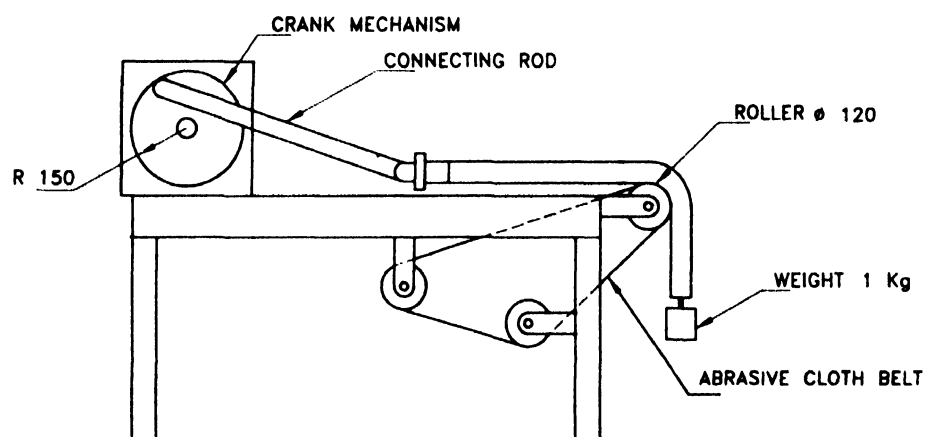
After the test, the hose shall withstand the electric strength test of **16.4** and shall not be damaged to such an extent that compliance with this standard is impaired.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows:

22.21 Addition

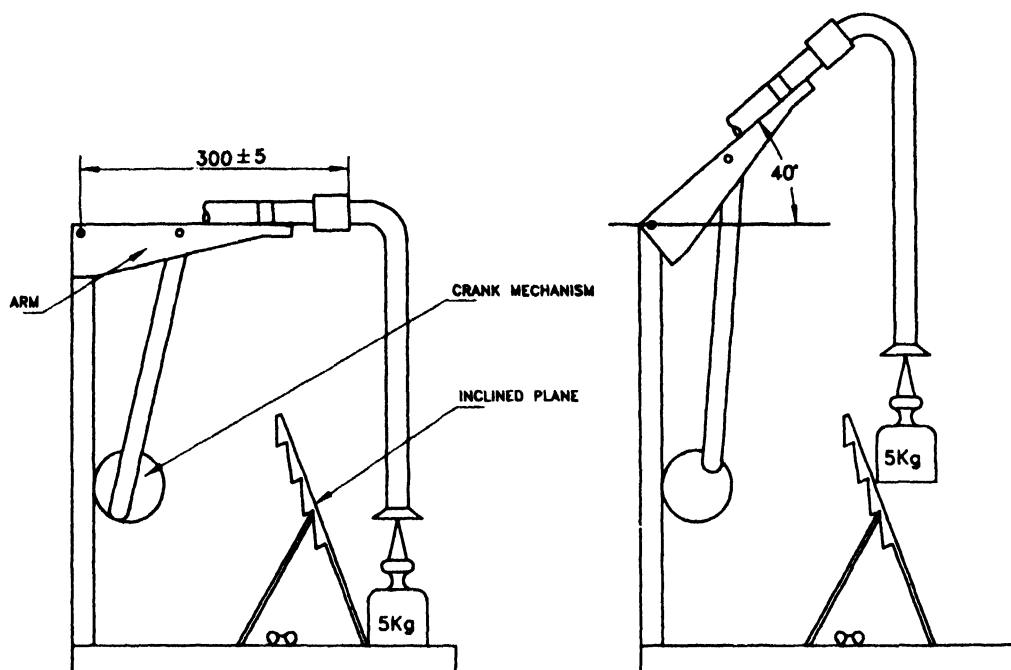
For water suction cleaning appliances, compliance



All dimensions in millimetres.

FIG. 101 APPARATUS FOR TESTING THE ABRASION RESISTANCE OF CURRENT — CARRYING HOSES

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All dimensions in millimetres.

FIG. 102 APPARATUS FOR TESTING THE RESISTANCE TO FLEXING OF CURRENT — CARRYING HOSES

with the requirement concerning protection against deposition of dirt has been adequately checked by the test of 15.3.

Vacuum cleaners shall be constructed so that the internal parts of the motors and electrical connections are not subjected to deposition of dust to the passage of air.

NOTE — This requirement is met if the air passes through the dust bag before it passes through the motor.

22.101 Additional subclauses

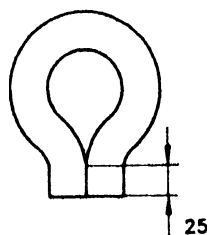
The appliances shall be fitted with a switch to control the motor for water suction cleaning appliances, the switch shall provide all-pole disconnection.

22.102 Water suction cleaning appliances shall not be provided with motorized cleaning heads. Compliance is checked by inspection.

22.103 Current-carrying hoses shall be resistant to cold conditions.

Compliance is checked by the following test.

A 600 mm length of hose is bent as shown in



All dimensions in millimetres.

FIG. 103 CONFIGURATION OF THE HOSE FOR THE FREEZING TREATMENT

Fig. 103 and the ends are tied together over a length of 25 mm.

The hose is then placed for 2 h in a cabinet having a temperature of $-15^{\circ}\text{C} \pm 2^{\circ}\text{C}$. Immediately after the hose is removed from the cabinet it is flexed three times, as shown in Fig. 104, at a rate of one flexing per second.

The test is carried out three times.

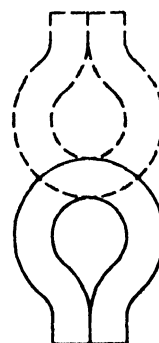
There shall be no cracks or breaks in the hose and it shall withstand the electric strength test of 16.4.

NOTE Any discolouration is neglected

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable.

INTERMEDIATE POSITION



POSITION OF THE HOSE AT START AND FINISH OF EACH FLEXING

All dimensions in millimetres.

FIG. 104 FLEXING POSITIONS FOR THE HOSE AFTER REMOVAL FROM THE FREEZING CABINET

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24 COMPONENTS

This clause of IS 302-1 (1979) is applicable.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1 (1979) is applicable.

25.1 Addition

Vacuum cleaners for animal grooming and water suction cleaning appliances shall be provided with a supply cord.

25.6 Addition

Instead of the types of supply cords specified the following applies.

Supply cords shall be not lighter than

- For hand-held appliances having a mass not exceeding 1.5 kg when fitted with the heaviest accessory but excluding the supply cord
- If rubber insulated, ordinary tough rubber sheathed flexible cord as per IS 9968 (Part 1) : 1988.
- If polyvinyl chloride insulated, light polyvinyl chloride sheathed flexible cord as per IS 694 : 1990.
- For Class III appliances:
 - If rubber insulated, ordinary tough rubber sheathed flexible cord as per IS 9968 (Part 1) : 1988.
 - If polyvinyl chloride insulated, flat twin flexible cord as per IS 694 : 1990.
- For other appliances:
 - If rubber insulated, ordinary tough rubber sheathed flexible cord as per IS 9968 (Part 1) : 1988
 - If polyvinyl chloride insulated, ordinary polyvinyl chloride sheathed flexible cord as per IS 694 : 1990.

25.13 Addition

Live conductors in a flexible hose shall have an insulation and sheath thickness at least equivalent to cord of $2 \times 0.75 \text{ mm}^2$ specified in IS 694 : 1990.

NOTE — The conductors used may consist of copper-plated steel wires.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1 (1979) is applicable

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable

29 CREEPAGE DISTANCES, CLEARANCES AND DISTANCES THROUGH INSULATION

This clause of IS 302-1 (1979) is applicable

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1 (1979) is applicable

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable

32 RADIATION HAZARDS

This clause of IS 302-1 (1979) is applicable

101 TESTS

101.0 Categories of Tests

Tests are classified as type, acceptance and routine tests.

101.1 Type Test

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the vacuum cleaners and water suction cleaning appliances shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

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Table 101 Schedule of Type Tests
(Clause 101.1)

Sl No.	Test	Clause Reference
i)	Protection against electric shock	8
ii)	Starting of motor operated appliances	9
iii)	Input	10
iv)	Temperature rise	11
v)	Electrical insulation and leakage current at operating temperature	13
vi)	Moisture resistance	15
vii)	Insulation resistance and electric strength (after humidity treatment)	16
viii)	Overload protection	17
ix)	Abnormal operation	19
x)	Stability and mechanical hazards	20
xi)	Mechanical strength	21
xii)	Construction	22
xiii)	Internal wiring	23
xiv)	Components	24
xv)	Cord grip and cord guard	25
xvi)	Terminals for external conductors	26
xvii)	Provision for earthing	27
xviii)	Screws and connections	28
xix)	Creepage distance and clearances	29
xx)	Resistance to heat fire and tracking	30
xxi)	Resistance to rusting	31

101.2 Acceptance Tests

The following shall constitute the acceptance tests

Test	Clause Reference
a) Protection against electric shock	8
b) Input	10
c) Temperature rise	11
d) Electrical insulation and leakage current at operating temperature	13
e) Moisture resistance	15
f) Insulation resistance and electric strength (after humidity treatment)	16
g) Provision for Earthing	27

NOTE For the purpose of acceptance tests, the humidity treatment shall be done for 24 h while conducting the test for moisture resistance (15).

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979)

101.3 Routine Test

The following shall constitute the routine tests

Test	Clause Reference
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Provision for earthing	27

ANNEX A

TABLES OF TYPE TEST

Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

SAMPLING PROCEDURE FOR ACCEPTANCE TESTS

Appendix B of IS 302-1 (1979) is applicable.

ANNEX C

ELECTRONIC CIRCUITS

Appendix C of IS 302-1 (1979) is applicable if electronic circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETERS

Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

Appendix E of IS 302-1 (1979) is applicable.

ANNEX F

IMPACT TEST APPARATUS

Appendix F of IS 302-1 (1979) is applicable.

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ANNEX G

THERMAL CONTROLS AND OVERLOAD RELEASES

Appendix G of IS 302-1 (1979) is applicable.

ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

Appendix H of IS 302-1 (1979) is applicable.

ANNEX J

TEST FOR FIRE RESISTING PROPERTIES

Appendix J of IS 302-1 (1979) is applicable

ANNEX K

BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKEL PLATING

Appendix K of IS 302-1 (1979) is applicable

ANNEX L

**APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL, STEEL AND
COPPER**

Appendix L of IS 302-1 (1979) is applicable.

ANNEX AA

(Foreword)

LIST OF REFERRED INDIAN STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No</i>	<i>Title</i>
302-1 (1979)	Safety of household and similar electrical appliances: Part 1 General requirements (<i>fifth revision</i>)	9968 (Part 1) : 1988	Elastomer insulated cables Part 1 For working voltages upto and including 1 100 V (<i>first revision</i>)
694 : 1990	PVC Insulated cables for working voltages upto and including 1 100 V (<i>third revision</i>)	11986 : 1987	Cotton backing cloth for abrasives

Bureau of Indian Standards

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Review of Indian Standards

Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Handbook' and 'Standards : Monthly Additions'.

This Indian Standard has been developed from Doc : No. ETD 32 (3932).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

BUREAU OF INDIAN STANDARDS

Headquarters:

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